

# Abdul Ahad

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Research interests: optimization & generalization theory in deep learning, adversarial robustness

## EDUCATION

### B.Tech, Computer Science and Engineering — Manipal Academy of Higher Education

2024 – 2028

CGPA: 9.50 / 10

Relevant coursework: Multivariable Calculus & Numerical Methods, Linear Algebra, Discrete Mathematics, Automata Theory & Compiler Design, DevOps & Cloud Computing

### B.S., Data Science and Applications — Indian Institute of Technology Madras

2024 – 2028

CGPA: 9.50 / 10

Relevant coursework: Deep Learning, Generative AI, Probability & Optimization, Algorithm Design, Machine Learning, Tools in Data Science, Application Development

## PUBLICATIONS

### [A Spectral Bound on Effective Sharpness for Fisher-Preconditioned Gradient Descent](#)

Abdul Ahad, Napa Lakshmi. Transactions on Machine Learning Research (TMLR), 2026. ISSN 2835-8856.

- Proved a general spectral bound on the effective sharpness seen by Fisher-preconditioned (natural-gradient) optimizers, and an alignment-aware Rayleigh analysis explaining when empirical-Fisher worst-case bounds are loose.
- Validated the bound on exact deep linear networks from 55 to 3,240 parameters, and extended it to matrix-free estimates on MNIST and CIFAR-10 ResNet-18 (11.2M parameters), where K-FAC reached 90.5% test accuracy.

## RESEARCH & TECHNICAL PROJECTS

### Universal Adversarial Cloak [\[GitHub\]](#) [\[HF Space\]](#) | PyTorch, Hugging Face, FaceNet, CLIP

- Designed a projected-gradient-descent (PGD) optimization engine that cloaks biometric and semantic identity against FaceNet and CLIP architectures, framed as a joint-loss adversarial optimization problem.
- Enforced strict  $L_\infty$ -norm constraints in the loss to keep perturbations human-imperceptible, achieving SSIM > 0.98 across all cloaked outputs.
- Ran an automated black-box transferability study against industry-standard oracle models (ArcFace, CLIP ViT-L/14) and used ablations to optimize Mean Residual Similarity (MRS).

### Geometric Image Reconstruction [\[GitHub\]](#) | PyTorch, NumPy, OpenCV

- Built a GPU-accelerated hill-climbing engine that reconstructs photographs from geometric primitives via a coarse-to-fine multi-stage schedule.
- Derived a closed-form analytic color solver and Sobel-based gradient routing, removing stochastic color search and improving convergence speed.
- Ran an ablation study across 11 architectural variants, reaching SSIM 0.6059 / PSNR 17.96 dB within a 1-minute compute budget — outperforming standard evolutionary baselines.

### Cryptex: Unsupervised Cryptanalysis Engine [\[GitHub\]](#) | Numba, MCMC, HMM

- Built a cryptanalysis suite for 7 classical cipher families combining adaptive MCMC with parallel tempering, Baum-Welch HMM/EM, and KPA-constrained solvers over Kneser-Ney smoothed 5-gram language models.
- Formulated a statistical cipher-type detector (Index of Coincidence, Kasiski examination, entropy) and a phase-transition analyzer mapping ciphertext length to decryption success, validating theoretical unicity distances empirically.
- Achieved 100% success and 0.0 Symbol Error Rate across a 130-run benchmark on substitution and transposition ciphers, accelerating the evaluation loop with Numba JIT-compiled tensor operations.

### Kivixa Workspace [\[GitHub\]](#) | Flutter, llama.cpp

- Built an offline, cross-platform workspace (distributed via F-Droid and winget) with on-device multi-model LLM inference and a sandboxed Model Context Protocol for secure autonomous file operations.
- Implemented an audio pipeline (Whisper STT, Kokoro neural TTS, voice-activity detection), a local vector database for semantic search, and a math engine for multivariable calculus and hypothesis testing.
- Implemented Git-like version control with SHA-256 content-addressable blob storage and automatic snapshots, plus a Lua-scriptable plugin architecture, a force-directed knowledge graph, and parallel state-driven orchestration.

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL

**Deep Learning & GenAI:** PyTorch, Transformers, Hugging Face, PEFT/LoRA, llama.cpp, Unsloth, LangGraph

**Research Methods:** MCMC & Bayesian sampling, hypothesis testing, ablation study design, statistical significance analysis

**Infrastructure:** Docker, FastAPI, PostgreSQL, MongoDB

## CERTIFICATIONS

[AWS Academy Graduate — Machine Learning](#): cloud-native AWS AI services (SageMaker, Comprehend, Fraud Detection).

[MLOps Specialization](#): Azure ML, MLFlow, Hugging Face inference pipelines.

[RAG & Agentic AI Professional](#): LangChain, LlamaIndex, multi-agent orchestration (CrewAI).